Machine Learning Foundations: Geometry, Optimization, and Linear Classification

From vectors and planes to loss functions, feature vectors, and classifiers

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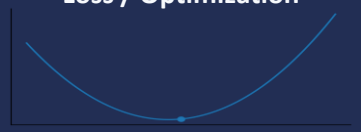
Point / Vector

$x=(3,2,2)$

Plane / Separator

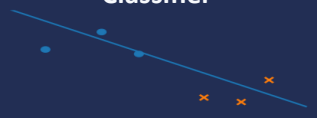
$3x_1+x_2-1=0$

Loss / Optimization



$\theta \leftarrow \theta-\gamma\nabla L(\theta)$

Classifier



$h(x)\in\{-1,+1\}$

Geometry gives us representation

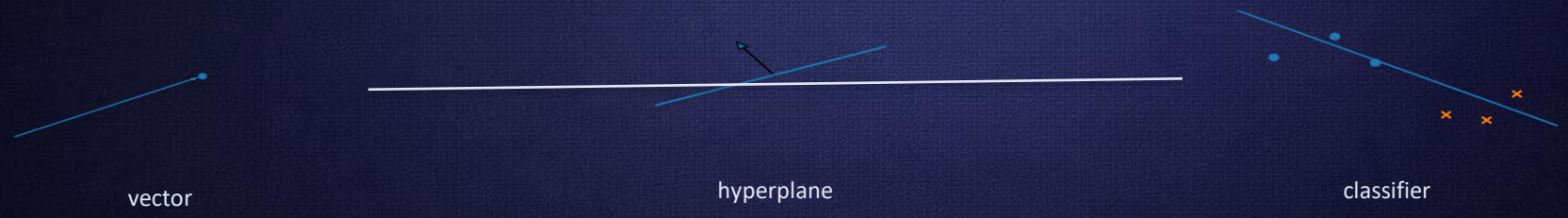
Optimization gives us learning

Classification gives us prediction

Foundations for early machine learning concepts and linear classification

Part I — Framing, Notation, and Roadmap

How the course’s mathematical language becomes machine learning language



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Why These Foundations Come First

Linear Algebra / Geometry

Probability / Uncertainty

Machine Learning

Calculus / Optimization

Computation / NumPy

Core idea

Machine learning is built from representation, geometry, optimization, uncertainty, and computation.

Why these prerequisites matter

- Vectors represent examples and parameters
- Planes represent decision boundaries
- Loss functions define prediction error
- Gradients tell us how to improve
- Probability helps us model uncertainty
- Computation makes the math executable

Before a model can learn, it needs representation, error measurement, and a method for improvement.

The Big Story: From Data to Decisions

Machine learning turns real-world objects into numerical representations, places them in a geometric space, and learns rules that predict labels or values.

Raw Object	Feature Vector	Point in Space	Model / Boundary	Prediction / Error
movie handwritten digit person record	encoded measurable properties of the object	vector becomes a location in geometry	score or separate regions of the space	output a label or value then compare to truth
raw input	$x \in R^d$	point / vector	$\theta \cdot x + \theta_0$	$h(x)$, $L(\theta)$

Running examples

movie → like/dislike • digit image → class label

Representation comes first: machine learning works by turning objects into vectors, placing them in a space, and learning decision rules over that space.